

Solutions for Linear Algebra: Step by step by
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Chapter 1

Linear Equations and Matrices

1.1 Exercises

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1.

- a) $x - y - z = 3$ is linear, each variable has a degree of 1
- b) $\sqrt{x} + y + z = 6$ is not linear, due to the square root of x.
- c) $\cos x + \sin y = 1$ is not linear, due to trigonometric functions for x and y
- d) $\exp x + y + z = 1$ is not linear, due to the exponential function
- e) $x - 2y + 5z = \sqrt{3}$ is linear, as the square root is not on one of the variables, but is on a constant value of 3.
- f) $x = -3y$ is linear
- g) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, is linear. No mention of y or z on the right hand side and it equates to a constant value.
- h) $\pi x + y + ez = 5$, is linear.
- i) $\sqrt{2}x + \frac{1}{2}y + z = 0$ is linear, the square root is just a coefficient of x
- j) $\sinh^{-1}(x) = \ln|x + \sqrt{x^2 + 1}|$ is not linear due to the trigonometric and logarithmic functions
- k) $\frac{\pi}{2}x - \sqrt{2}y + z \sin \pi = 0$ is linear. $\sin \pi$ is a constant coefficient of z, same goes for y and x. each variable has a degree of 1
- l) $x^{2^0} + y^{3^0} + z^{3^0} = 0$ is linear. The trick is each power resolves to 1, as 2^0 and 3^0 is 1
- m) $y^{\cos^2(x) + \sin^2(x)} + x - z = 9$. This is linear, trick is to know that $\cos^2(x) + \sin^2(x)$ is a trig identity, resolving to 1

2.

a)

$$x + y = 2 \quad (1)$$

$$x - y = 0 \quad (2)$$

$$\begin{array}{r} x + y = 2 \\ + (x - y = 0) \\ \hline 2x = 2 \end{array}$$

$$x = 1$$

$$y = 1$$

b)

$$2x - 3y = 5$$

$$x - y = 2$$

$$2x - 2y = 4$$

$$\begin{array}{r} 2x - 3y = 5 \\ -(2x - 2y = 4) \\ \hline -y = 1 \end{array}$$

$$y = -1$$

$$x = 1$$

c)

$$2x - 3y = 35$$

$$x - y = 2$$

$$2x - 2y = 4$$

$$\begin{array}{r} 2x - 3y = 35 \\ -(2x - 2y = 4) \\ \hline -y = 31 \end{array}$$

$$y = -31$$

$$x = -29$$

d)

$$\begin{array}{rcl}
 5x - 7y & = & 2 \\
 9x - 3y & = & 6 \\
 3x - y & = & 2 \\
 15x - 5y & = & 10 \\
 15x - 21y & = & 6 \\
 & & 15x - 5y = 10 \\
 & & \underline{-(15x - 21y = 6)} \\
 & & 16y = 4
 \end{array}$$

$$\begin{array}{rcl}
 4y & = & 1 \\
 y & = & \frac{1}{4} \\
 3x = 2 + \frac{1}{4} & = & \frac{9}{4} \\
 x & = & \frac{3}{4}
 \end{array}$$

e)

$$\begin{array}{rcl}
 \pi x - 5y & = & 2 \\
 \pi x - y & = & 1 \\
 & & \pi x - 5y = 2 \\
 & & \underline{-(\pi x - y = 1)} \\
 & & -4y = 1 \\
 y & = & -\frac{1}{4} \\
 \pi x = 1 + y & = & \frac{3}{4} \\
 x & = & \frac{3}{4\pi}
 \end{array}$$

f)

$$\begin{array}{rcl}
 ex - ey & = & 2 \\
 ex + ey & = & 0 \\
 & & ex - ey = 2 \\
 & & \underline{+(ex + ey = 0)} \\
 & & 2ex = 2 \\
 x & = & \frac{1}{e} \\
 y & = & -\frac{1}{e}
 \end{array}$$